

ROHAN PATEL

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SUMMARY

Data Scientist with a strong foundation in machine learning, NLP, and computer vision, specializing in Data Engineering. Proficient in Python, SQL, Tableau, and advanced tools like LangChain and HuggingFace, with a proven ability to transform data into actionable insights and deploy high-impact models.

SKILLS

Programming: Python, MS SQL, PostgreSQL, MySQL, R, Java, Hadoop, Spark, .NET, Angular, Javascript

Skills: Machine Learning, Deep Learning, Programming, Database, Data Scraping, Data Wrangling, Computer Vision, Natural Language Processing NLP, Large Language Models LLMs, Recommender Systems, Sentiment Analysis, Time Series Forecasting, Data Visualization, Tensorflow, Keras, Pytorch, Pandas, Numpy, Version Control, Communication

Tools: Anaconda, AWS, SSMS, Jupyter Notebook, Tableau, Git, MATLAB, pgAdmin, PowerBI, Excel, VSCode

WORK EXPERIENCE

Data Engineer, Meditab Software Pvt Ltd: Jan'19 – Jan'22 (3yrs 1m)

- Coordinated ETL driven data migrations and conversion for healthcare organizations like IPAs, TPAs, MSOs, and ACOs from their legacy systems to MSSQL Server, seamlessly transferring over 5M patient records and 200K provider profiles, improving database speed by 35% and enhancing security by 20%.
- Managed 2.5 TB of sensitive healthcare data, ensuring HIPAA compliance. Implemented anonymization for 20M patient records and over 1M provider profiles for claims adjudication, authorizations, and case management modules.
- Trained and mentored 3 team members on SQL, backup and recovery processes, stored procedures, and underlying database structure accelerating their onboarding by 30% and ensuring seamless collaboration on critical projects.
- Conducted data analysis to develop 25+ custom dashboards / client to provide crucial insights to save \$2M annually.
- Implemented robust data backup of 1.5 TB of critical data achieving 99.9% recovery success rate.
- Engineered automated custom data importing, processing 80+ files monthly achieving very high accuracy of 95%.
- Designed database applications such as interfaces, data transfer mechanisms, global temporary tables, data partitions, stored procedures and function-based indexes to enable efficient access of the generic database structure

PROJECT EXPERIENCE

ChatPDF: Retrieval-Augmented Generation (RAG), **Generative AI**, LLM, LangChain, HuggingFace, Groq | [Code](#) | [App](#)

- Engineered a state-of-the-art PDF QA system for users to query multiple PDFs using Langchain, and Huggingface
- Designed and implemented a RAG pipeline with history-aware retrievers, boosting response accuracy by 30%
- Developed an interactive chat interface with session-based history tracking allowing seamless navigation between new and older chats enhancing user experience

CorpFinChat AI: Qdrant, LangChain, OpenAI, EDGAR API, RAG, **GenAI**, Web Scraping | [Code](#) | [App](#)

- Built financial chatbot, processing 10-K filings of almost 10k companies with EDGAR API integration
- Created a RAG pipeline with Qdrant, OpenAI, Groq, LangChain and Streamlit cutting retrieval latency by 40%, memory usage by 20%, session-based tracking and concurrent user management
- Integrated SEC filings API and web scraping for real-time data scraping like stock prices and financial news

DeepGuard: Computer Vision, Deep Learning, CNNs, Classification, Data Augmentation | [Code](#) | [App](#)

- Steered the development of deepfake image detection system leveraging CNNs achieving 85% detection accuracy.
- Applied advanced data preprocessing and augmentation techniques leading to a 15% increase in detection accuracy and a 20% reduction in false positives.
- Deployed a real-time detection model via Streamlit achieving a sub-200ms processing time per image

RecipeRover: Recommender Systems, Machine Learning, Data Preprocessing, Web Scraping | [Code](#) | [App](#)

- Spearheaded the development of an end-to-end recipe recommendation system, encompassing web scraping of over 7500 recipes, meticulous data cleaning, preprocessing, modeling, and web application deployment
- Implemented a content-based recommendation system with TF-IDF and cosine similarity to suggest similar recipes tailored to user preferences, achieving an average recommendation speed of under 100 milliseconds.
- Orchestrated the deployment as a user-friendly web application using Streamlit, successfully attracting engagement from a targeted user base of over 50 users within the initial launch phase

MovieLens: SQL, PostgreSQL, SQLAlchemy, Streamlit, Data Visualization, Query Optimization | [Code](#)

- Analyzed the MovieLens dataset containing 20M records to identify and optimize long-running queries.
- Implemented indexing and normalization optimizations to improve performance ranging from 40% to 90%.
- Deployed a robust web application named Movie Mania using Streamlit and SQLAlchemy integrating data visualizations and movie trailers for immersive user experience.
- Identified and fine-tuned resource hungry and frequent queries to streamline retrieval and enhance responsiveness.

Customer & Sales KPI Dashboard: Tableau, Data Visualization | [Sales](#) | [Customer](#)

- Dynamic filtering by region, year, and category to compare annual performance across different years.
- Calculated fields to derive profit ratio, sales ratio, YoY growth, and customer profitability
- Key metrics like total sales, profit, orders, and customer segments using KPIs, stacked bar and line graphs.
- Best practices in dashboard design including clean layouts, consistent color schemes, and guided storytelling.

Amazon Prime Dashboard: Tableau, Data Visualization | [Dashboard](#)

- Designed a dashboard analyzing Amazon Prime show trends, customer demographics, and viewing behavior.
- Utilized maps, radial bar chart, pie chart, area graphs for top ratings, region-wise shows, and show types.
- Implemented filters and parameters to enable interactive storytelling and flexible data exploration.

ReviewRadar: Sentiment Analysis, NLP, spaCy, NLTK, Data Cleaning, Word Cloud | [Code](#)

- Developed a sentiment analysis model using NLP techniques for a massive dataset of 11 million reviews. Achieved 88.5% accuracy on test data using Support Vector Machines
- Implemented data cleaning and pre-processing techniques including tokenization, stop words removal, stemming, lemmatization and text normalization with spaCy.
- Generated a word cloud visualization to communicate sentiment trends within the reviews effectively

HeroClassifier: Computer Vision, Deep Learning, Transfer Learning, VGG16, Image Classification | [Code](#) | [App](#)

- Orchestrated a deep learning project for 10 class image classification, showcasing proficiency in data scraping, preprocessing, and model deployment
- Utilized CNN and transfer learning with the VGG16, achieving an impressive 88.67% accuracy on unseen data
- Implemented a user-friendly interface leveraging Streamlit for model deployment, enabling users to upload superhero images and receive predictions instantly

Reducing Telecom Customer Churn: Customer Retention, Predictive Modelling, SVC, Python | [Code](#)

- Unveiled key drivers of telecom churn to anticipate customer defection and implement targeted retention strategies
- Built a Support Vector Classifier on the Customer Churn data to predict customer churn with an accuracy of 81%
- Analyzed data to collect information used to build targeted customer retention tactics thus lowering client churn

Exasens Analysis: R, Data Cleaning, EDA, Clustering, Bagging, Machine Learning, Data Visualization | [Code](#)

- Conducted a comprehensive analysis of the Exasens dataset using R programming involving rigorous data cleaning and exploratory data analysis (EDA).
- Experimented clustering algorithms including DBSCAN, K-Means, Hierarchical Clustering, SVC, Bagging, Random Forest, Ridge Classifier to finalize ensemble method of bagging with decision tree with 88% accuracy.
- Created insightful visualizations using tidyverse, ggplot2, and dplyr to communicate findings and enhance data-driven decision-making.

Multivariate Oil Sales Forecasting: Time Series Forecasting, R, Data Imputation, Statistical Modeling | [Code](#)

- Developed a forecasting model to predict oil sales prices from a non-stationary time series using R performing extensive EDA and data cleaning.
- Evaluated multiple models including ETS, Holt-Winters and Auto ARIMA with the Auto ARIMA model outperforming others based on RMSE and MAE metrics.
- Conducted ADF tests for stationarity, Mann-Kendall and Hurst-Exponent methods to identify Seasonality and Trend. Performed Outlier removal using residual fitted plots and Cooke's distance method.

Airfoil Self-Noise: R Programming, Regression Analysis, Machine Learning, Random Forest | [Code](#)

- Engaged in Regression Problem of 1.5K instances and 6 attributes leveraging R programming
- Performed Exploratory data analysis like scaling, and checking outliers along with scatter, box, and pair plots
- Executed various ML models including Multi-linear Regression, Non-Linear Regression, Ridge Regression, Lasso Regression, Boosting, Bagging, Random Forest Classifier, Bayesian Additive Regression Trees (BART)
- Recorded the best performance by the Random Forest with a training error of 12% on the dataset

Pandas from Scratch: Data Manipulation, Python, Pandas

- Programmed Pandas functionality from the ground up, utilizing fundamental Python data structures
- Engineered Pandas DataFrame functionality encompassing column manipulation (create, modify or delete), mean computation, dataframe slicing, type conversion (astype), data selection (loc), and index reconfiguration (set_index)
- Facilitated in gaining a comprehensive understanding of Pandas' core functionalities for data manipulation and cleaning

SunriseTransformers: Full-Stack Development, MSSQL, .NET Framework, Angular, Deployment, Hosting | [App](#)

- Coded and deployed a website using the latest technologies for a client to boost the reach of the company by 30%
- Employed MSSQL for managing and storing various products, services offered, and any inquiries of any buyer
- Integrated backend coding using .NET technology with MS SQL to attain smooth data flow from DB to the front-end
- Implemented typescript for integrating with .NET API to achieve a quick page loading time of under 2 seconds for each page

EDUCATION

Masters: Data Science - 3.9/4, University at Buffalo, The State University of New York, May 2024

Bachelors: Computer Engineering - 8.74/10, GTU, India, May 2019

CERTIFICATIONS

- Google Data Analytics Professional Certificate, June 2022 - Acquired Skills (Data cleaning, Data Analysis, & Visualization) and mastered Tools (SQL, R programming, Tableau, Spreadsheets)
- Data Science using Python Workshop, January 2018 - Skills (Data Science Workflow, Data Manipulation with Pandas, Data Visualization, Data Preprocessing, ML Python Libraries like NumPy and SciPy)